Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



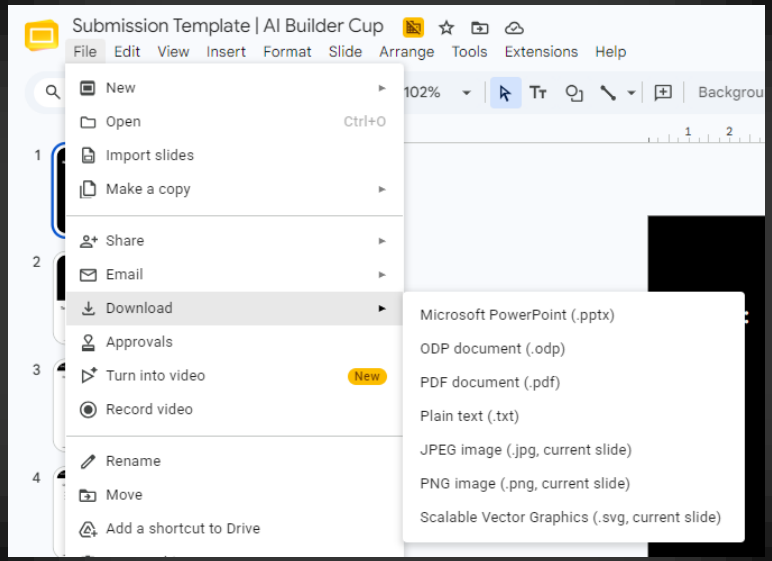
Team Details

Team name: Agent Ram

Team leader: Ramasamy P

Track: Sustainability · Google Cloud AI Builder Cup 2026 (JAPAC)

Problem: Operators receive millimetre-scale InSAR motion, but not a ranked, consequence-aware shortlist of which 250 m



TerraSentinel AI

Which 250 metres should an engineer inspect today?

Satellites measure motion. TerraSentinel ranks consequence. Gemini drafts the ticket — every number from tools, never invented.

Live · BigQuery · Cloud Run · labelled synthetic demo data

Brief about the idea

- Idea: turn InSAR point clouds into a consequence-ranked Act list for critical infrastructure.
- Maths (Python / BigQuery SQL) compute speed, acceleration, progress, RPI, tier and ETA bands.
- Gemini reasons and drafts inspection tickets from tool results only — a number guard rejects ungrounded figures.
- Rain amplifies hazard; it never creates a tier alone. USGS earthquakes are context only.
- Outlook = time-to-threshold band if today's trend continues — never a collapse / failure date.
- Shipped: FastAPI + React/MapLibre on Cloud Run, optional BigQuery warehouse, Data Hub validate-then-import.
- Honesty: bundled corridors are synthetic demonstration data, labelled in the UI.

Opportunities

- **How different is it from any of the other existing ideas?**
- **How will it be able to solve the problem?**
- **USP of the proposed solution**

Differentiation

- Not another InSAR viewer
- 250 m chainage sites + RPI
- Consequence × hazard × confidence
- Pass-to-pass alerts (diffs)
- Gemini narrates; engine decides

How it solves it

- Ingest → locate → measure
- Classify + WEAK data guard
- Rank Act-today shortlist
- Outlook band (const + accel)
- Work order with verified numbers

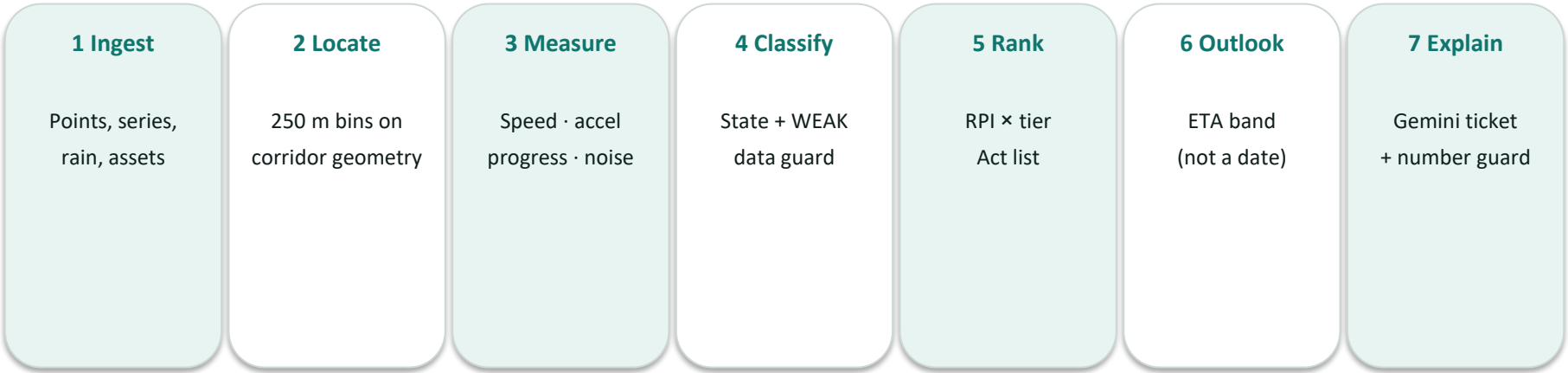
USP

- Honest ETA bands, not dates
- Number-guarded GenAI tickets
- Same maths: local or BigQuery
- Production posture on GCP
- Judge can sign in and act

List of features offered by the solution

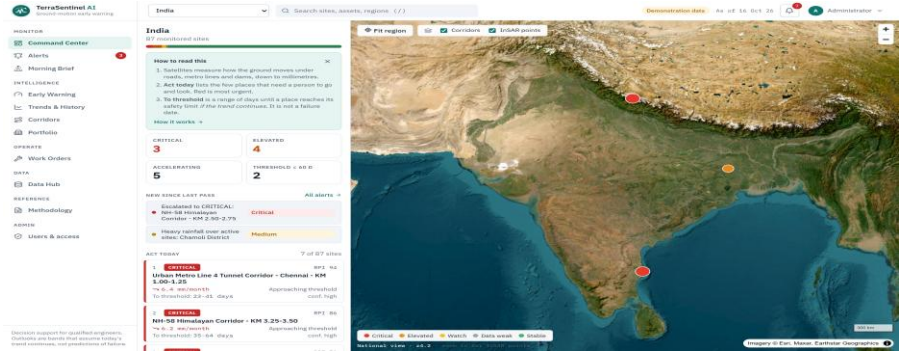
- Command Center — map, KPIs, Act today list (CRITICAL / ELEVATED ranked by consequence)
- Site drawer — RPI, speed, rain trigger, outlook band, trend across satellite passes
- Alerts — what changed between passes (not a duplicate of the Act list)
- Morning Brief · Early Warning · Trends · Corridors · Portfolio
- Work Orders — template-first ticket; optional Gemini polish; tool-grounded numbers
- Data Hub — validate-then-import (CSV/GeoJSON), rainfall fetch, provenance, BigQuery sync, reset to demo
- Access — admin / analyst / viewer RBAC, session auth, audit log
- Methodology page — published formulae so every score is reproducible

Process flow diagram or Use-case diagram

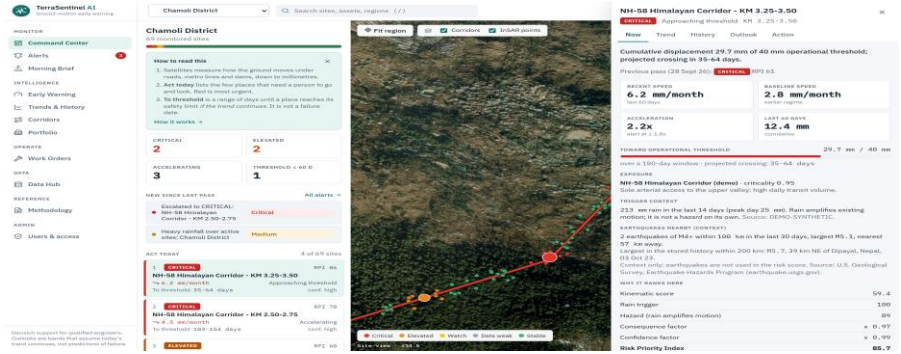


Use-case: Corridor engineer opens Command Center → selects CRITICAL NH-58 KM 3.25–3.50 → reads 35–64 day threshold band if trend continues → generates inspection work order.

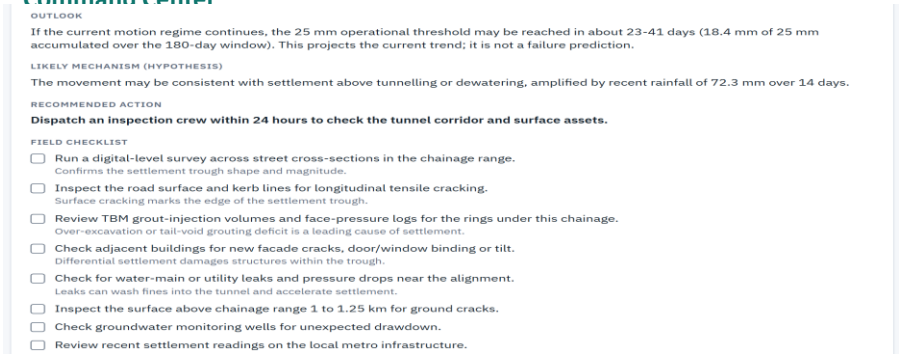
Wireframes/Mock diagrams of the proposed solution (optional)



Command Center



Site · NH-58



OUTLOOK

If the current motion regime continues, the 25 mm operational threshold may be reached in about 23-41 days (18.4 mm of 25 mm accumulated over the 180-day window). This projects the current trend; it is not a failure prediction.

LIKELY MECHANISM (HYPOTHESIS)

The movement may be consistent with settlement above tunnelling or dewatering, amplified by recent rainfall of 72.3 mm over 14 days.

RECOMMENDED ACTION

Dispatch an inspection crew within 24 hours to check the tunnel corridor and surface assets.

FIELD CHECKLIST

- [ ] Run a digital-level survey across street cross-sections in the chainage range.
  Confirms the settlement trough shape and magnitude.
- [ ] Inspect the road surface and kerb lines for longitudinal tensile cracking.
  Surface cracking marks the edge of the settlement trough.
- [ ] Review TBM grout-injection volumes and face-pressure logs for the rings under this chainage.
  Over-excavation or tail-void grouting deficit is a leading cause of settlement.
- [ ] Check adjacent buildings for new facade cracks, door/window binding or tilt.
  Differential settlement damages structures within the trough.
- [ ] Check for water-main or utility leaks and pressure drops near the alignment.
  Leaks can wash fines into the tunnel and accelerate settlement.
- [ ] Inspect the surface above chainage range 1 to 1.25 km for ground cracks.
- [ ] Check groundwater monitoring wells for unexpected drawdown.
- [ ] Review recent settlement readings on the local metro infrastructure.

Work Order



Data Hub

Architecture diagram of the proposed solution

L5 Edge — Cloud Run · HTTPS · CSP · CSRF · rate limits · sessions · GCS state

L4 Experience — React SPA · MapLibre · React Query · RBAC UI

L3 API — FastAPI · auth · audit · SSRF-safe fetch · Data Hub ingest

L2 Intelligence — Detection engine · RPI · tiers · ETA bands · alerts

L1 GenAI — Gemini tool-calling · number guard · template fallback

L0 Data — Local workspace XOR BigQuery · USGS public context (never scores)

Technologies to be used in the solution

Google Cloud

- Cloud Run (max-instances=1)
- Cloud Build + Artifact Registry
- BigQuery (partitioned InSAR)
- Secret Manager
- Cloud Storage (state volume)
- Gemini API (work orders)

Application

- Python 3.12 · FastAPI
- Deterministic detection engine
- React · Vite · MapLibre
- TanStack Query
- Session auth + RBAC
- Validate-then-import Data Hub

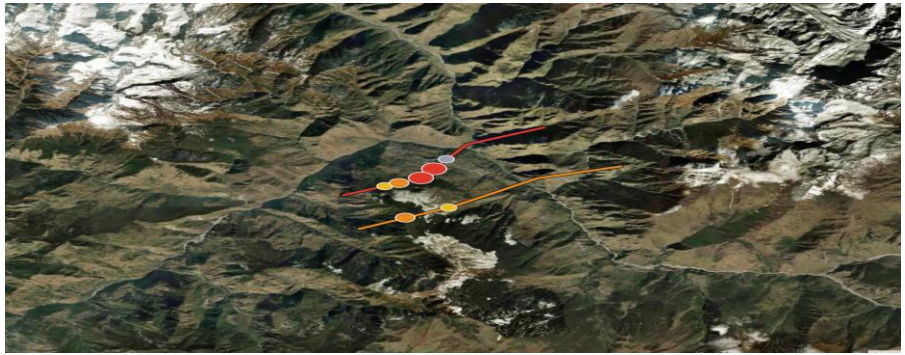
Data & integrity

- Synthetic demo (labelled)
- Open-Meteo rainfall fetch
- USGS seismicity (context)
- Number guard on Gemini
- Provenance registry
- pytest + security checklist

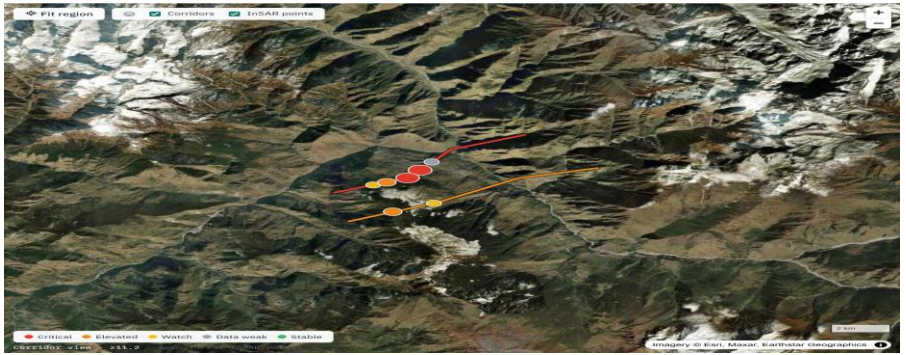
Estimated implementation cost (optional)

- Cup demo posture (kept warm): Cloud Run 1 vCPU / 2 GiB, min-instances=1 during judging — small continuous cost.
- BigQuery: demo dataset ~62k time-series rows; queries cached in-process (TTL); hard max-bytes-billed guard.
- Gemini Flash: work-order narratives only; template fallback if quota/latency — no GenAI on the hot ranking path.
- Secret Manager + GCS state: pennies at demo scale.
- Scale path: raise instances, materialize point-assignment, keep engine deterministic so GenAI spend stays optional.
- Estimated Cup-month ops (order of magnitude): low tens of USD if kept warm; near-zero if scaled to zero off-hours.

Snapshots of the prototype



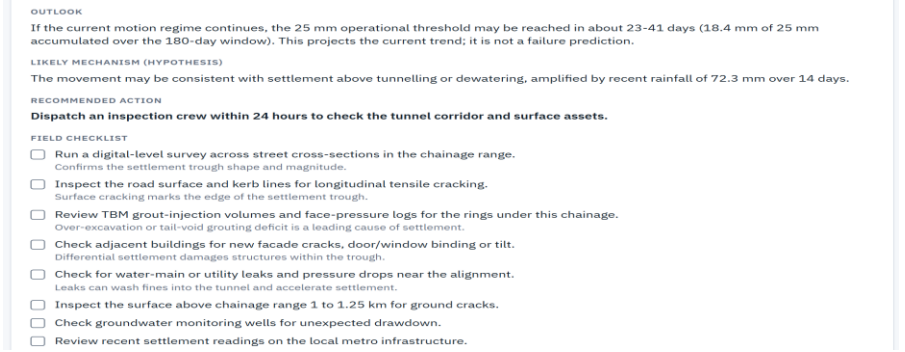
Map · corridor risk



Chamoli · ranked sites



Site Now · RPI / outlook



Gemini work order

Prototype Performance report/Benchmarking

- Demo corpus: 1,355 scatterers · ~62,330 displacement rows · 87 sites of 250 m · 7 CRITICAL/ELEVATED on latest pass.
- Worked site NH-58 KM 3.25–3.50: RPI 86 · ~6.2 mm/month · outlook band 35–64 days if trend continues.
- Integrity: GenAI never computes RPI/tier/ETA; template path always available; synthetic data labelled.
- Latency (live Cloud Run + BigQuery, warm): region summary ~0.4 s · sites ~0.3 s · alerts ~2 s (after hot-path prefetch).
- Cold start mitigated: startup prefetch of BigQuery frames + current rankings before historical warm.
- Security: CSRF header, cookie sessions, rate limits, RBAC, audit events, no secrets in git.
- Tests: pytest suite covering engine, ingest, auth hardening, BigQuery fake runner.

Additional Details/Future Development (if any)

- Connect production InSAR providers (same schema; Data Hub validate-then-import already in place).
- Materialize spatial point-assignment in BigQuery for national-scale corridors.
- Multi-region HA and horizontal Cloud Run when session store moves off single-process files.
- Operator workflows: assign/escalate SLAs, mobile field checklist, offline packet export.
- Deeper rain + geology covariates as amplifiers only — never sole drivers of tier.
- Keep the non-negotiable: outlook bands ≠ failure dates; GenAI narrates tools only.

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

Submission links

GitHub public repository

https://github.com/ramasamyp-aiengineer/terra-sentinel-ai

Demo video (3 minutes)

See repo: docs/pitch/TerraSentinel-AI-Demo-3min.mp4 (upload to Drive/YouTube for the portal)

Final product (live)

https://terrasentinel-598899050820.asia-southeast1.run.app

Sign-in: username + password are shown on the login page (judge demo hint).

Judge walkthrough: docs/JUDGE.md in the repository.

Honesty rules (non-negotiable)

Outlooks are time-to-threshold bands under a continuing-regime assumption — not collapse predictions.

Gemini narrates tool numbers only; it does not invent scores, ETAs or tiers.

Demo / synthetic data is labelled in the product.

Secrets stay in .env / Secret Manager — never in git.

Google Cloud H2S

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Build for the world.

Thank you

Thank you — Team Agent Ram

TerraSentinel AI · Google Cloud AI Builder Cup 2026 · Sustainability

Live demo + GitHub + this deck · Ask us to walk Command Center → Work Order → Data Hub.

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